Explainable Deep Learning for Phishing Website Detection: A Compact Feature Representation with Global and Local Interpretability via SHAP and LIME

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Abstract*—Phishing websites remain a major cybersecurity threat, and although deep learning models achieve strong detection accuracy, they often lack transparency, while explainable machine learning studies mostly use classical models and provide only global feature importance. This study proposes an explainable deep learning framework for the detection of phishing websites using a large structured dataset of 88,647 samples and 111 input features derived from the network, domain, WHOIS, SSL, and URL. After preprocessing, Random Forest-based feature selection, and correlation-based reduction (111→98→47→27), a compact set of 27 features was obtained while maintaining or improving predictive performance. A Multilayer Perceptron (MLP) was developed on this feature set and compared against five conventional models : Logistic Regression, SVM, Decision Tree, XGBoost, and a Class-Weighted Random Forest using held-out testing, 10-fold cross-validation, and paired t-test and Wilcoxon signed-rank tests. The Class-Weighted Random Forest achieved the strongest performance among the evaluated benchmark models, with 97.06% accuracy and a 95.81% F1-score, while the proposed MLP provided a competitive deep learning approach, achieving 96.32% accuracy, 94.73% F1-score, and 99.10% ROC-AUC, together with integrated explainability. For additional deep learning comparison, TabNet, FT-Transformer, and Tabular ResNet were also evaluated on the same compact feature representation. SHAP and LIME were applied to the MLP to provide both global feature-importance rankings and instance-level explanations, addressing a gap where deep learning models on structured phishing datasets are rarely paired with dual-level explainability. Results show that domain activation time, URL length, and the number of dots in the domain were the most influential features. The proposed framework demonstrates that a compact, interpretable, and carefully validated deep learning approach can achieve competitive phishing detection performance while offering transparent and interpretable explanations for its predictions.

***Index Terms*—Phishing Detection, Explainable AI, SHAP, LIME, Deep Learning, Multilayer Perceptron, Feature Selection**

I. Introduction

A. Background

Phishing remains one of the most damaging cybersecurity threats, with the Anti-Phishing Working Group (APWG) reporting approximately 3.8 million phishing attacks in 2025. These attacks continue to cause significant financial losses and compromise sensitive personal and organizational information worldwide. As phishing attackers use more advanced techniques, traditional methods based on fixed rules and keywords are no longer effective enough. Therefore, researchers are increasingly using Machine Learning and Deep Learning (DL) approaches for phishing detection [6], [10], [14]. While these models achieve high accuracy, their black-box nature limits trust among security experts, making Explainable Artificial Intelligence (XAI) techniques such as SHAP and LIME in-

creasingly important for building transparent, high-performing phishing detection systems [1], [2].

B. Problem Statement

Although several studies combine deep learning with XAI, the integration remains limited: high-performing models are rarely designed with interpretability in mind, while interpretable studies mostly rely on simpler ML models rather than deep learning. Most existing work reports only global feature importance, omitting instance-level explanations that are crucial for analysts, and often uses limited validation without isolating the contribution of individual system components. This highlights the need for a phishing detection framework that combines strong predictive performance, reliable explanations, and thorough evaluation.

C. Summary of Research Gap

A review of prior work shows that feature selection, explainable machine learning, and deep learning for phishing detection have mostly been studied separately: structured datasets are typically handled either by deep learning models without explainability or by traditional ML paired with SHAP/LIME, and comprehensive evaluation protocols combining baseline comparison, cross-validation, and statistical testing remain rare. Section II reviews this literature in detail and identifies the specific gap this study addresses.

D. Contributions

To address these gaps, this study proposes an explainable deep learning framework for phishing website detection using a large structured dataset of 88,647 instances and 111 input features (112 columns including the binary target) covering domain, WHOIS, SSL, and URL-based information, with a Multilayer Perceptron (MLP) as the primary classifier, explained via SHAP and LIME; traditional ML algorithms are used only as performance comparisons rather than direct competitors. The main contributions are:

- An explainable deep learning framework using an MLP model with SHAP and LIME for global and local phishing-prediction explanations.
- A compact 27-feature representation, reduced from 111 initial features by removing constant features, applying Random Forest importance ranking, and using correlation-based clustering.
- Detailed comparison of the MLP against five conventional machine learning models using held-out testing, 10-fold cross-validation, and statistical significance testing.
- Comparative evaluation of three supplementary deep learning models (TabNet, FT-Transformer, Tabular ResNet) and an exploratory MLP-Tabular ResNet hybrid.
- A combined performance-and-explainability analysis, presenting the MLP as an explainable model instead of the most accurate model.

The remainder of the paper is organized as follows: Section II reviews related literature, Section III describes the proposed methodology, Section IV presents experimental results, and Section V concludes with future research directions.

II. Literature Review

Deep learning methods have shown promising results in phishing detection [6], [10], [14], while multimodal and LLM-based approaches have introduced additional detection strategies [4], [5], [15]. Table I summarizes seven representative studies highlighting the major limitations identified in the literature.

The remaining studies follow three recurring patterns. Some employ deep learning or hybrid models mainly for accuracy without formal explainability, including CNN/LSTM URL classifiers [6] and DL-versus-quantized-LLM comparisons [7], [13], [14]. Others combine classical ML, ensembles, or transformers with SHAP, LIME, or feature ranking but exclude deep learning and mainly provide global explanations [1], [2], [3], [8], [11], [12]. Multimodal, federated, and LLM-assisted approaches introduce additional data types or privacy methods but often require extra computational or communication costs and rely on qualitative explanations [4], [5], [9], [10].

Overall, three gaps remain: limited quantitative global and local explainability for deep learning-based phishing detection [1], [2], [6], [8], [14]; limited combined SHAP-LIME analysis of individual predictions [1], [2], [12]; and insufficient joint evaluation of multiple ML/DL models, statistical significance, and feature reduction. This study addresses these gaps through dual SHAP-LIME analysis of the proposed MLP, combined model evaluation, and compact feature representation.

III. Proposed Methodology

A. Dataset Description

This study uses the publicly available Phishing Websites Dataset (Vrbančič, 2020, Mendeley Data, DOI 10.17632/72ptz43s9v.1), containing 88,647 website instances (58,000 legitimate, 30,647 phishing) and 112 columns: 111 URL-, domain-, WHOIS/DNS-, SSL/TLS-, and redirect-based features plus a binary target 'phishing' (0 = legitimate, 1 = phishing). The dataset was chosen for its comprehensive feature coverage and prior use in explainability-focused phishing research. Fig. 1 summarizes the full processing pipeline, from the raw dataset through preprocessing,feature selection, correlation-based reduction, train-test splitting, model training and evaluation, to the final explainable AI analysis; feature selection and correlation-based reduction were performed on the full cleaned dataset before the train-test split, which is discussed as a limitation in Section V-B.

B. Data Preprocessing

The raw dataset was checked for missing values, duplicate rows, conflicting labels, and constant features. No missing values were found. First, 13 conflicting observations (7 legitimate and 6 phishing) were removed. Then, 1,437 duplicate rows were removed, leaving 87,197 samples. Next, 13 constant features were removed because they did not provide useful information. This reduced the features from 111 to 98, resulting in a cleaned dataset with 87,197 samples, 98 input features, and one target variable.

TABLE I
COMPARISON OF REPRESENTATIVE STUDIES AND OUR CONTRIBUTION

Ref.	Data & Method	Key Limitation	Our Contribution
[2]	∼500K multi-source URLs; RF with SHAP, LIME, PDP, and permutation importance.	XAI focused on feature selection, not DL.	Global and local XAI for a DL classifier.
[3]	Structured phishing dataset; explainable ML framework.	Traditional ML only.	Explainable DL using SHAP and LIME.
[4]	∼3.2M URLs + MTLP; CatBoost, CNN1D, CodeBERT, EfficientNet-B7.	Complex multimodal architecture; high computational cost.	Simpler DL model with global and local XAI.
[6]	ISCX-URL2016 (20K); CNN, LSTM, and CNN-LSTM.	No explainability; static URL features only.	SHAP and LIME for DL explanations.
[7]	83,446 phishing emails; ML, DL, and quantized LLMs.	No quantitative XAI; mainly LLM-based explanations.	Quantitative SHAP and LIME with DL.
[12]	108,078 URLs; ML, ensembles, neural nets, transformers + SHAP/LIME.	SHAP analysis excluded DL; mainly global explanations.	SHAP and LIME directly on DL with global/local explanations.
[15]	Mendeley + TR-OP; PhishDebate multi-agent LLM.	Qualitative LLM explanations; smaller dataset.	Quantitative SHAP/LIME on a large structured dataset.

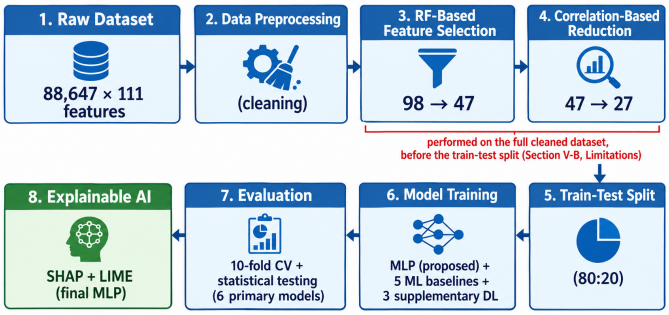


Fig. 1. Overall proposed framework, from the raw dataset to explainable AI.

C. Feature Selection

A Random Forest model trained on the full cleaned dataset (87,197 samples, before the train-test split) ordered the 98 features by importance, and cumulative importance was computed: 32 features represented for about 80% of total importance, 40 features for about 90%, and a 95% cumulative-importance threshold selected 47 features, which were used for correlation-based reduction. Performing feature selection before the train-test split is acknowledged as a methodological limitation in Section V.

D. Correlation-Based Feature Reduction

Using the 47 selected features, a correlation matrix identified 189 feature pairs with absolute correlation $\geq$ 0.90. Representing features as graph nodes connected by high-correlation edges showed three correlated clusters (sizes 2, 21, and 2); for each cluster, the feature with the highest Random Forest importance was kept as the representative while its highly correlated related features were removed. This reduced the 21-feature cluster to three representatives (`qty_plus_directory`, `qty_asterisk_directory`, `qty_at_directory`) and the two smaller clusters to one representative each, while non-clustered features were kept unchanged resulting in a final set of 27 features covering URL structure, directory/file characteristics, domain and DNS information, redirect behavior, and SSL/TLS properties, used for all later modeling.

E. Train-Test Split

The 27-feature dataset was split 80:20 into training (69,757 samples: 45,364 legitimate, 24,393 phishing) and testing (17,440 samples: 11,342 legitimate, 6,098 phishing) sets, preserving the original class distribution; the test set was reserved solely for final evaluation.

F. Baseline Machine Learning Models

Five conventional machine learning models : Logistic Regression, SVM , Decision Tree , XGBoost , and Class-Weighted Random Forest were trained on the same 27-feature training data and evaluated using Accuracy, Precision, Recall, F1-score, and ROC-AUC to compare the proposed MLP.

G. Class Imbalance Handling

With the dataset moderately imbalanced (65.03% legitimate vs. 34.97% phishing), two handling strategies were compared: class weighting applied to a Random Forest model, and SMOTE oversampling applied only to the training set, which balanced the training data from 45,364/24,393 to 45,364/45,364 samples (90,728 total) while leaving the test set unchanged to avoid data leakage. Both approaches were compared across the five evaluation metrics to help later model optimization.

H. Random Forest Hyperparameter Optimization

RandomizedSearchCV was used to tune the Random Forest over `n_estimators` (100–399), `max_depth` (None, 10–30), `min_samples_split` (2–9), `min_samples_leaf` (1–4), `max_features` (*sqrt*, *log2*), and `class_weight` (*balanced*, *balanced_subsample*), optimizing F1-score. The best configuration — 152 trees, max_depth=25, log2 features, min_samples_leaf=1, min_samples_split=3, and balanced subsample weighting — achieved a cross-validation F1-score of 0.9559. However, the tuned model did not outperform the Class-Weighted Random Forest in terms of held-out test accuracy. Therefore, the Class-Weighted Random Forest with the better test performance was retained as the final baseline for later comparisons.

I. Proposed MLP Model and Hyperparameter Optimization

The primary proposed model is a Multilayer Perceptron (MLP) trained using the 27 standardized (StandardScaler) features. Seven MLP configurations with different hidden layers, neuron numbers, learning rates, and L2 regularization values were evaluated on the held-out test set. The best configuration used two hidden layers with 128 and 64 neurons, ReLU activation, the Adam optimizer, a learning rate of 0.001, L2 regularization = 0.0001, batch size 32, and early stopping. This configuration was selected as the final MLP for comparison with the baseline models. The final MLP was then evaluated using 10-fold cross-validation to check its stability (Section IV-D).

J. Additional Deep Learning Models

For broader comparison, three supplementary tabular deep learning models : TabNet (attention-based), FT-Transformer (transformer-based), and Tabular ResNet (residual) were trained and evaluated on the same 27-feature data under 10-fold cross-validation using the same five metrics, though they were not used in the SHAP/LIME explainability analysis.

K. Additional Hybrid Deep Learning Experiment

An experimental hybrid architecture combining MLP and Tabular ResNet branches before the final classification layer was also developed and evaluated on the same 27-feature representation and evaluation setup, to test whether combining different learned features improves phishing detection performance.

L. Model Evaluation and Statistical Testing

The proposed MLP was compared with the five baseline models (six 'main' models) using Accuracy, Precision, Recall, F1-score, and ROC-AUC under 10-fold cross-validation; the three supplementary DL models were evaluated separately and excluded from the main statistical analysis. Both the paired t-test and the Wilcoxon signed-rank test were considered for pairwise comparison, but given only ten paired folds and uncertain normality, the Wilcoxon signed-rank test ($\alpha = 0.05$) was adopted as the primary significance test, alongside a confusion matrix for the final MLP and a feature-ablation comparison of the MLP trained on the 47-feature versus 27-feature representations.

M. Explainable AI: SHAP and LIME

To ensure interpretability, SHAP and LIME were applied to the final MLP at both global and local levels: SHAP quantified overall feature importance across the dataset and explained two individual predictions (one correctly classified phishing site and one correctly classified legitimate site), while LIME was applied to the same two instances to provide complementary local explanations together offering a comprehensive global-and-local view of model behavior.

IV. Experimental Results and Discussion

A. Feature Selection and 47 vs. 27 Feature Comparison

Random Forest-based feature selection and correlation-based reduction (Sections III-C and III-D) reduced the feature set from 98 to 47 and finally to 27 features covering URL structure, domain/DNS/WHOIS properties, SSL/TLS information, and redirect behavior. To verify that this reduction did not compromise performance, the MLP was evaluated using both feature sets. The 27-feature model achieved higher accuracy (0.9632 vs. 0.9592), recall (0.9454 vs. 0.9310), F1-score (0.9473 vs. 0.9411), and ROC-AUC (0.9910 vs. 0.9906) than the 47-feature model, with only a marginal decrease in precision (0.9493 vs. 0.9514). These results indicate that the removed 20 features contributed little additional predictive value, supporting the effectiveness of the compact 27-feature representation.

B. Traditional ML Baseline and Class-Imbalance Results

Five conventional machine learning models were evaluated on the 27-feature test set (results reported with the MLP in Table IV, Section IV-E). Logistic Regression scored lowest (92.87% accuracy, 89.97% F1-score), while SVM and Decision Tree performed similarly (F1 93%). The ensemble methods performed best: XGBoost reached 96.39% accuracy and 94.85% F1-score, while the Class-Weighted Random Forest achieved the strongest overall performance (97.06% accuracy, 95.81% F1-score). For the moderately imbalanced dataset (Section III-G), class weighting slightly outperformed SMOTE (97.06% vs. 96.96%). Hyperparameter tuning was also explored but did not improve upon the Class-Weighted Random Forest, which was kept as the final baseline.

C. Proposed MLP: Hyperparameter Tuning and Final Performance

Seven MLP configurations with different architectures, learning rates, regularization values, and iteration limits were evaluated (Table II). MLP_4, with two hidden layers of 128 and 64 neurons, achieved the highest accuracy (96.32%), recall (94.54%), F1-score (94.73%), and ROC-AUC (99.10%). Although MLP_5 achieved slightly higher precision (95.22%), its lower recall and F1-score made MLP_4 the better overall choice. Thus, MLP_4 was selected as the final architecture and evaluated using 10-fold stratified cross-validation for stability assessment.

TABLE II
MLP Hyperparameter Configuration Results

Model	Layers	LR	Alpha	Max Iter.	Acc.	Prec.	Rec.	F1	AUC
MLP_4	**(128,64)**	**0.0010**	**0.0001**	32	**0.9632**	0.9493	**0.9454**	**0.9473**	**0.9910**
MLP_5	(128,64,32)	0.0010	0.0001	30	0.9616	**0.9522**	0.9374	0.9447	0.9903
MLP_3	(64,32)	0.0010	0.0001	34	0.9606	0.9468	0.9401	0.9435	0.9907
MLP_7	(64,32)	0.0010	0.0010	44	0.9604	0.9459	0.9405	0.9432	0.9909
MLP_6	(64,32)	0.0005	0.0001	46	0.9583	0.9474	0.9326	0.9399	0.9906
MLP_2	(64,)	0.0010	0.0001	48	0.9575	0.9444	0.9333	0.9388	0.9905
MLP_1	(32,)	0.0010	0.0001	40	0.9558	0.9393	0.9339	0.9366	0.9892

D. Supplementary Deep Learning Models and Hybrid Experiment

TabNet, FT-Transformer, and Tabular ResNet were evaluated alongside the MLP's 10-fold cross-validation average (Table III, distinct from its held-out test result in Table IV); Tabular ResNet performed best, while TabNet lagged behind. These models were used only for supplementary comparison and excluded from statistical significance and SHAP/LIME analyses. An MLP–Tabular ResNet hybrid achieved 96.14% accuracy and 99.21% ROC-AUC, but did not outperform the Class-Weighted Random Forest and was therefore not adopted.

TABLE III
PRIMARY MLP VS. SUPPLEMENTARY DEEP LEARNING MODELS

Model	Acc.	Prec.	Rec.	F1	AUC
MLP (Primary)	0.9607	0.9458	0.9416	0.9436	0.9919
TabNet	0.9505	0.9264	0.9326	0.9295	0.9888
FT-Transformer	0.9591	0.9435	0.9397	0.9414	0.9925
Tabular ResNet	**0.9637**	**0.9472**	**0.9491**	**0.9481**	**0.9933**

E. Cross-Validation, Statistical Significance, and Final Comparison

The six main models were evaluated using 10-fold cross-validation and the held-out test set (Table IV). Class-Weighted Random Forest achieved the best overall performance, followed by XGBoost and MLP, with low cross-validation variance (accuracy std ≤ 0.003). Wilcoxon signed-rank testing ($\alpha = 0.05$) confirmed significant differences between Random Forest and both MLP and XGBoost, and between MLP and the other baselines across all metrics, except MLP vs. XGBoost precision ($p = 0.492$).

TABLE IV
FINAL COMPARISON OF MAIN MODELS ON THE HELD-OUT TEST SET

Model	Acc.	Prec.	Rec.	F1	AUC
Logistic Regression	0.9287	0.8859	0.9139	0.8997	0.9770
SVM	0.9541	0.9376	0.9308	0.9342	0.9873
Decision Tree	0.9525	0.9325	0.9316	0.9321	0.9477
XGBoost	0.9639	0.9463	0.9508	0.9485	0.9930
MLP	0.9632	0.9493	0.9454	0.9473	0.9910
Class-Weighted RF	**0.9706**	**0.9565**	**0.9597**	**0.9581**	**0.9946**

F. Explainability: SHAP and LIME Analysis

SHAP was used to compute global feature importance across the dataset (Fig. 2) and to explain two representative individual predictions. Domain activation time, URL length, and the number of dots in the domain were the three most influential features overall, followed by several URL-structure indicators (slashes, at-signs, asterisks).

For a correctly classified phishing website, the SHAP explanation (Fig. 3) shows that a high hyphen count in the URL was the dominant driver toward 'phishing,' followed by domain activation time, dot count in the domain, file length, MX-server count, URL length, and the at-sign count in the directory; only the dot count in the URL, directory length, and most notably

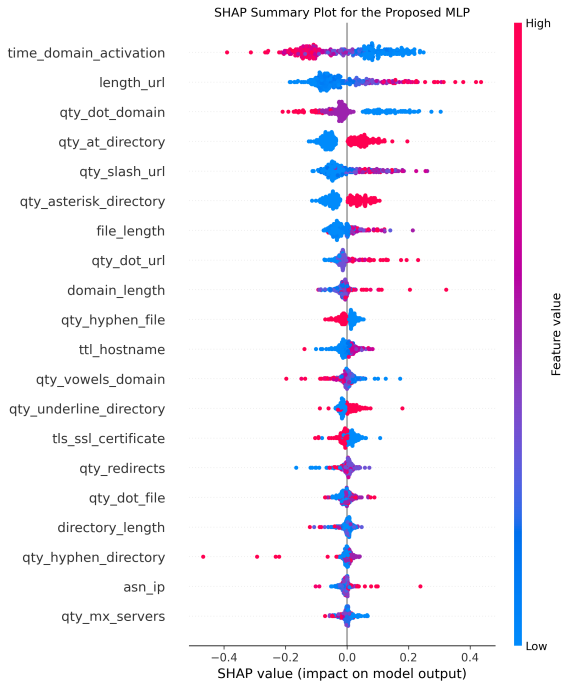


Fig. 2. SHAP summary plot showing the global feature contributions of the proposed MLP.

the hyphen count in the directory pulled the prediction back toward 'legitimate.'

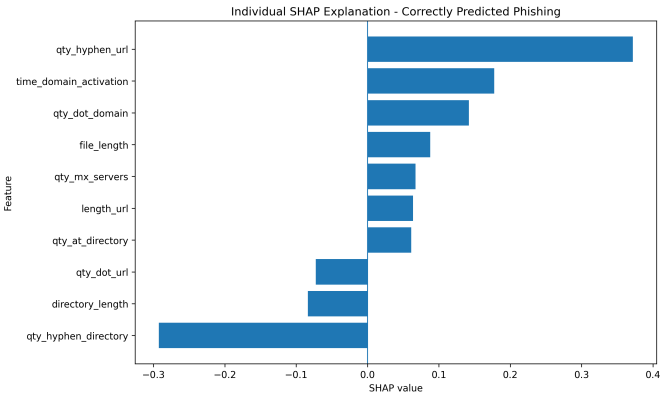


Fig. 3. Local SHAP explanation for a correctly predicted phishing website.

LIME, applied to the same correctly classified phishing website (predicted probability 0.971), identified a long URL and a low domain-dot count as the two strongest indicators pushing the prediction toward 'phishing' (Fig. 4), while a high hyphen count in the file path and a low dot count in the URL pulled weakly toward 'legitimate' reinforcing that the MLP's decisions draw on a mix of URL, domain, directory, and network-related patterns rather than any single rule.

G. Error Analysis and Discussion

The final MLP achieved a 3.68% overall misclassification rate (641/17,440), with a higher false-negative rate (5.46%)

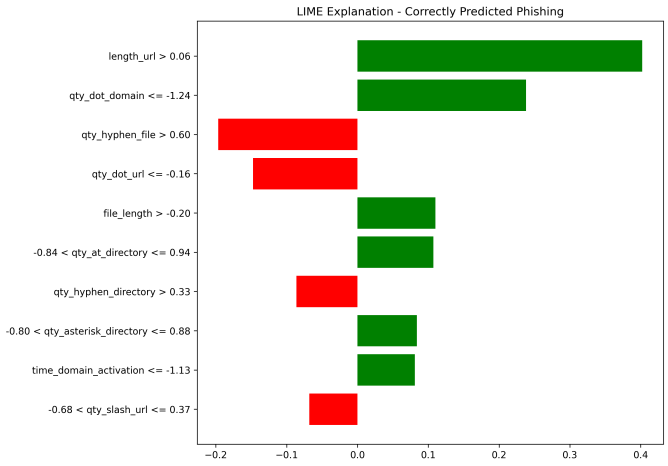


Fig. 4. LIME explanation for a correctly predicted phishing website.

than false-positive rate (2.72%). This asymmetry supports emphasizing recall alongside accuracy for phishing detection. Although the Class-Weighted Random Forest achieved the best raw performance, the MLP remained competitive while providing detailed global and local SHAP-LIME explanations.

TABLE V
CONFUSION MATRIX OF THE FINAL MLP (TEST SET)

	Predicted Legitimate	**Predicted Phishing**
Actual Legitimate	11,034	308
Actual Phishing	333	5,765

V. CONCLUSION AND FUTURE RESEARCH DIRECTIONS

A. Conclusion

This study proposed an explainable deep learning framework for phishing detection using a compact 27-feature representation derived from an original 112-column dataset containing 88,647 samples, including 111 input features and one target variable. Reducing the feature set from 47 to 27 did not degrade MLP performance and improved several metrics. Among the six primary models, the Class-Weighted Random Forest achieved the best raw performance (97.06% accuracy, 95.81% F1-score), while the proposed MLP remained competitive with 96.32% accuracy, 94.73% F1-score, and 99.10% ROC-AUC. Supplementary deep learning models and an MLP-Tabular ResNet hybrid did not surpass the Random Forest. The main contribution is the integration of a competitive MLP with dual SHAP-LIME explainability, providing both global and local interpretations for structured phishing detection.

B. Limitations

Feature selection and correlation-based reduction were performed before the train-test split, introducing a potential risk of information leakage. The test set was also used to select among the seven MLP configurations (Section III-I), introducing a risk of selection bias. The study also relies on a single 2020 dataset, limiting generalization to newer phishing attacks. In addition, the dataset contains only engineered URL, domain, WHOIS, and SSL/TLS features, excluding raw HTML and webpage screenshots. Explainability is limited to representative local predictions, while global SHAP analysis is performed across the test set.

C. Future Research Directions

Future work should perform feature selection and correlation reduction using only training data, preferably within nested cross-validation, and evaluate the framework on newer and additional phishing datasets. Incorporating HTML content and webpage screenshots could improve detection of visually deceptive attacks. Further studies should explore explicit feature interactions, dedicated tuning of supplementary deep learning models, broader instance-level explainability, and real-time deployment to assess explanation stability, prediction speed, and robustness against evolving phishing tactics.

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